Setup des machines GOAD

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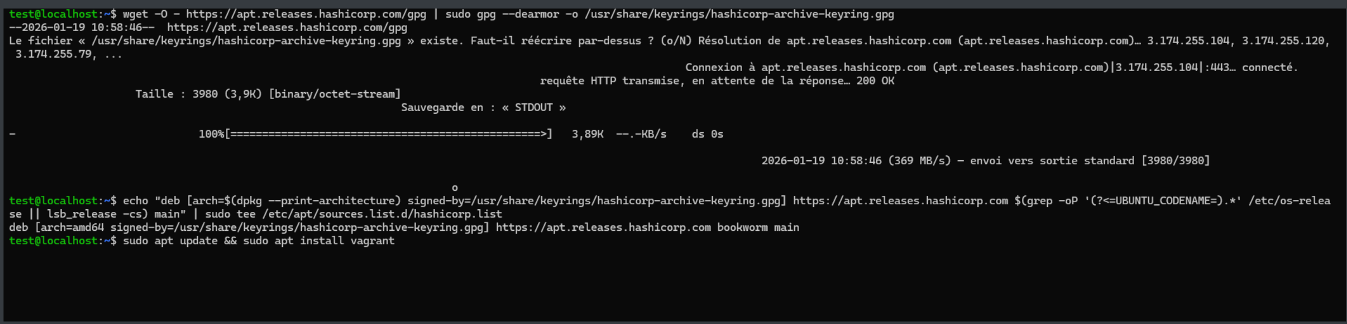
o/ Mise en place de l'hôte (source)

Pour commencer, nous allons mettre en place notre machine Linux pour pouvoir installer correctements les différentes machines. On commence par installer Vagrant sur le serveur selon la procédure https://developer.hashicorp.com/vagrant/install#linux :

```
wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com $(grep -oP '(?<=UBUNTU_CODENAME=).*' /etc/os-release || lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install vagrant
```



```
test@localhost:~$ wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
--2026-01-19 10:58:46--  https://apt.releases.hashicorp.com/gpg
Le fichier « /usr/share/keyrings/hashicorp-archive-keyring.gpg » existe. Faut-il réécrire par-dessus ? (o/N) Résolution de apt.releases.hashicorp.com (apt.releases.hashicorp.com)… 3.174.255.104, 3.174.255.120,
 3.174.255.79, ...
                                                     Connexion à apt.releases.hashicorp.com (apt.releases.hashicorp.com)|3.174.255.104|:443… connecté.
                                            requête HTTP transmise, en attente de la réponse… 200 OK
          Taille : 3980 (3,9K) [binary/octet-stream]
                                  Sauvegarde en : « STDOUT »

-                     100%[==================================================>]   3,89K  --.-KB/s    ds 0s

                                                        2026-01-19 10:58:46 (369 MB/s) - envoi vers sortie standard [3980/3980]

                                       o
test@localhost:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com $(grep -oP '(?<=UBUNTU_CODENAME=).*' /etc/os-relea
se || lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
deb [arch=amd64 signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com bookworm main
test@localhost:~$ sudo apt update && sudo apt install vagrant
```

Ensuite on installe les plugins Vagrant nécessaires :

```
vagrant plugin install vagrant-reload vagrant-vbguest winrm winrm-fs winrm-elevated
```

```
test@localhost:~$ vagrant plugin install vagrant-reload vagrant-vbguest winrm winrm-fs winrm-elevated
Installing the 'vagrant-reload' plugin. This can take a few minutes...
Fetching virtualbox-0.8.6.gem
Fetching vagrant-reload-0.0.1.gem
Installed the plugin 'vagrant-reload (0.0.1)'!
Installing the 'vagrant-vbguest' plugin. This can take a few minutes...
Fetching micromachine-3.0.0.gem
Fetching vagrant-vbguest-0.32.0.gem
Installed the plugin 'vagrant-vbguest (0.32.0)'!
Installing the 'winrm' plugin. This can take a few minutes...
Installed the plugin 'winrm (2.3.9)'!
Installing the 'winrm-fs' plugin. This can take a few minutes...
Installed the plugin 'winrm-fs (1.3.5)'!
Installing the 'winrm-elevated' plugin. This can take a few minutes...
Installed the plugin 'winrm-elevated (1.2.3)'!
```

Ensuite d'après la doc nous devons vérifier la version actuelle de Python et instller le paquet venv correspondant :

```
python3 --version
sudo apt install python3.11-venv
```

```
test@localhost:~$ python3 --version
Python 3.11.2
test@localhost:~$ sudo apt install python3.11-venv
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  augeas-lenses db-util guestfish guestfs-tools guestmount hfsplus icoutils ldmtool libafflib0v5
  libarchive-tools libaugeas0 libbfio1 libconfig9 libdate-manip-perl libewf2 libgrpc29
  libguestfs-hfsplus libguestfs-perl libguestfs-reiserfs libguestfs-tools libguestfs-xfs libguestfs0
  libhfsp0 libhivex0 libldm-1.0-0 libprotoc32 libre2-9 librpm9 librpmio9 libstring-shellquote-perl
  libsys-virt-perl libtsk19 libvhdi1 libvmdk1 libwin-hivex-perl libyara9 lsscsi nfs-kernel-server
  racc rpm-common ruby-addressable ruby-bcrypt-pbkdf ruby-builder ruby-childprocess ruby-concurrent
  ruby-diffy ruby-ed25519 ruby-erubi ruby-excon ruby-faraday ruby-ffi ruby-fog-core ruby-fog-json
  ruby-fog-libvirt ruby-fog-xml ruby-formatador ruby-google-protobuf
  ruby-googleapis-common-protos-types ruby-googleauth ruby-grpc ruby-i18n ruby-jwt ruby-libvirt
  ruby-listen ruby-little-plugger ruby-log4r ruby-logging ruby-memoist ruby-mime-types
  ruby-mime-types-data ruby-mini-portile2 ruby-multi-json ruby-multipart-post ruby-net-scp
  ruby-net-sftp ruby-net-ssh ruby-nokogiri ruby-oj ruby-os ruby-pkg-config ruby-public-suffix
  ruby-rb-inotify ruby-ruby2-keywords ruby-signet ruby-vagrant-cloud ruby-xml-simple ruby-zip scrub
  sleuthkit supermin syslinux vagrant-libvirt virt-p2v zerofree
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
Les paquets supplémentaires suivants seront installés :
  libpython3.11 libpython3.11-dev libpython3.11-minimal libpython3.11-stdlib python3.11
  python3.11-dev python3.11-minimal
Paquets suggérés :
  python3.11-doc binfmt-support
Les paquets suivants seront mis à jour :
  libpython3.11 libpython3.11-dev libpython3.11-minimal libpython3.11-stdlib python3.11
  python3.11-dev python3.11-minimal python3.11-venv
8 mis à jour, 0 nouvellement installés, 0 à enlever et 371 non mis à jour.
Il est nécessaire de prendre 12,6 Mo dans les archives.
Après cette opération, 10,2 ko d'espace disque supplémentaires seront utilisés.
```

1/ Mise en place des VMs GOADyo

Ensuite, on va cloner le repo GOAD et on va s'y déplacer :

```
git clone https://github.com/Orange-Cyberdefense/GOAD.git
cd GOAD
```

```
test@localhost:~$ git clone https://github.com/Orange-Cyberdefense/GOAD.git
cd GOAD
Clonage dans 'GOAD'...
remote: Enumerating objects: 7258, done.
remote: Counting objects: 100% (1448/1448), done.
remote: Compressing objects: 100% (410/410), done.
remote: Total 7258 (delta 1186), reused 1038 (delta 1038), pack-reused 5810 (from 1)
Réception d'objets: 100% (7258/7258), 24.71 Mio | 3.11 Mio/s, fait.
Résolution des deltas: 100% (3601/3601), fait.
test@localhost:~/GOAD$
```

Ensuite on va lancer le script `goad.sh` qui permet de vérifier les prérequis, créer le venv python et installer les dep Ansible :

```
./goad.sh
```

```
test@localhost:~/GOAD$ ./goad.sh
Python version in use : 3.11.2
python version >= 3.8 ok
venv module is installed. continue
[+] venv not found, start python venv creation
Requirement already satisfied: pip in /home/test/.goad/.venv/lib/python3.11/site-packages (23.0.1)
Collecting pip
  Downloading pip-25.3-py3-none-any.whl (1.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 8.8 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 23.0.1
    Uninstalling pip-23.0.1:
      Successfully uninstalled pip-23.0.1
Successfully installed pip-25.3
Collecting rich (from -r requirements_311.yml (line 1))
  Downloading rich-14.2.0-py3-none-any.whl.metadata (18 kB)
Collecting psutil (from -r requirements_311.yml (line 2))
  Downloading psutil-7.2.1-cp36-abi3-manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.
whl.metadata (22 kB)
Collecting Jinja2 (from -r requirements_311.yml (line 3))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting pyyaml (from -r requirements_311.yml (line 4))
  Downloading pyyaml-6.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_6
4.whl.metadata (2.4 kB)
Requirement already satisfied: setuptools in /home/test/.goad/.venv/lib/python3.11/site-packages (from
 -r requirements_311.yml (line 7)) (66.1.1)
Collecting ansible_runner (from -r requirements_311.yml (line 8))
  Downloading ansible_runner-2.4.2-py3-none-any.whl.metadata (3.2 kB)
Collecting ansible-core==2.18.0 (from -r requirements_311.yml (line 11))
  Downloading ansible_core-2.18.0-py3-none-any.whl.metadata (7.7 kB)
Collecting pywinrm (from -r requirements_311.yml (line 12))
  Downloading pywinrm-0.5.0-py3-none-any.whl.metadata (11 kB)
Collecting azure-identity (from -r requirements_311.yml (line 14))
  Downloading azure_identity-1.25.1-py3-none-any.whl.metadata (88 kB)
Collecting azure-mgmt-compute (from -r requirements_311.yml (line 15))
  Downloading azure_mgmt_compute-37.1.0-py3-none-any.whl.metadata (81 kB)
Collecting azure-mgmt-network (from -r requirements_311.yml (line 16))
  Downloading azure_mgmt_network-30.1.0-py3-none-any.whl.metadata (96 kB)
Collecting boto3 (from -r requirements_311.yml (line 18))
  Downloading boto3-1.42.30-py3-none-any.whl.metadata (6.8 kB)
Collecting proxmoxer (from -r requirements_311.yml (line 20))
  Downloading proxmoxer-2.2.0-py3-none-any.whl.metadata (5.2 kB)
Collecting requests (from -r requirements_311.yml (line 21))
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
community.general:12.2.0 was installed successfully
/home/test/GOAD

   _____   _____         _____
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  \_____| \_|||_/________\_____/
    Game Of Active Directory
      Pwning is coming

Goad management console type help or ? to list commands

[*] goad config file not found, create file /home/test/.goad/goad.ini
[*] Start Loading default instance
[*] lab instances :
[-] No instance found, change your config and use install to create a lab instance

GOAD/vmware/local/192.168.56.X > |
```

Une fois que tout est bon, on est arrivé dans la console de GOAD, et on peut installer les machines pour Virtualbox :

```
exit
```

Et de retour dans le terminal bash on fait :

```
# On commence par installer la machine GOAD Ansible
sudo docker build -t goadansible .
# Et on installe le tout
./goad.sh -t install -l GOAD -p virtualbox -m docker
```

```
test@localhost:~/GOAD$ ./goad.sh -t install -l GOAD -p virtualbox -m docker

   _____   _____          _____
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  \_____| \_|||_/_/      \_\_____/
    Game Of Active Directory
      Pwning is coming

Goad management console type help or ? to list commands

[*] lab instances :
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
9ffafb-goad-vmware	GOAD	vmware	192.168.56.0/24	not provided	Yes	

```
[+] Current Settings
[*] Current Lab         : GOAD
[*] Current Provider    : virtualbox
[*] Current Provisioner : docker
[*] Current IP range    : 192.168.56.X
[*] Extension(s)        :

Create lab with theses settings ? (y/N)y
[*] Create instance folder
[*] Create instance providing files
[*] Instance vagrantfile created : /workspace/3fc86a-goad-virtualbox/provider/Vagrantfile
[*] Create lab provisioning file inventory_disable_vagrant
[+] Lab inventory file created : /workspace/3fc86a-goad-virtualbox/inventory_disable_vagrant
[*] Create instance provisioning files
[+] Instance inventory file created : /workspace/3fc86a-goad-virtualbox/inventory
[*] Create instance extensions inventory files
[*] Instance 3fc86a-goad-virtualbox created
[*] Current user is in docker group
[*] CWD: /
[*] Running command :  docker images |grep -c "goadansible"
0
[-] Docker image goadansible not found
[*] You should build the docker image with : "sudo docker build -t goadansible ."
[+] Instance 3fc86a-goad-virtualbox loaded
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
> 3fc86a-goad-virtua…	GOAD	virtualbox	192.168.56.0/24	not provided	No	

Je risque d'avoir une erreur de desync :

```
TASK [vulns/adcs_esc6 : Configure ATTRIBUTESUBJECTALTNAME2 on CA - ESC6] *****************************
An exception occurred during task execution. To see the full traceback, use -vvv. The error was:    at System.Management.Automation.Interpreter.EnterTryCatchFinallyInstruction.Run(InterpretedFrame frame)
fatal: [srv03]: FAILED! => {"changed": false, "msg": "internal error: failed to become user 'essos.local\\administrator': Exception calling \"CreateProcessAsUser\" with \"9\" argument(s): \"The trust relations
hip between this workstation and the primary domain failed.\r\n\""}
 [started TASK: include_role : ps on dc01]
 [started TASK: include_role : ps on dc02]
 [started TASK: include_role : ps on dc03]

TASK [ps : Play task ../ad/GOAD/scripts/asrep_roasting2.ps1] *****************************************
changed: [dc03]

PLAY RECAP *******************************************************************************************
dc01                       : ok=11   changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
dc02                       : ok=16   changed=10   unreachable=0    failed=0    skipped=6    rescued=0    ignored=0
dc03                       : ok=10   changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv02                      : ok=13   changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
srv03                      : ok=10   changed=1    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

[-] 3 fails abort.
[-] Something wrong during the provisioning task : vulnerabilities.yml
```

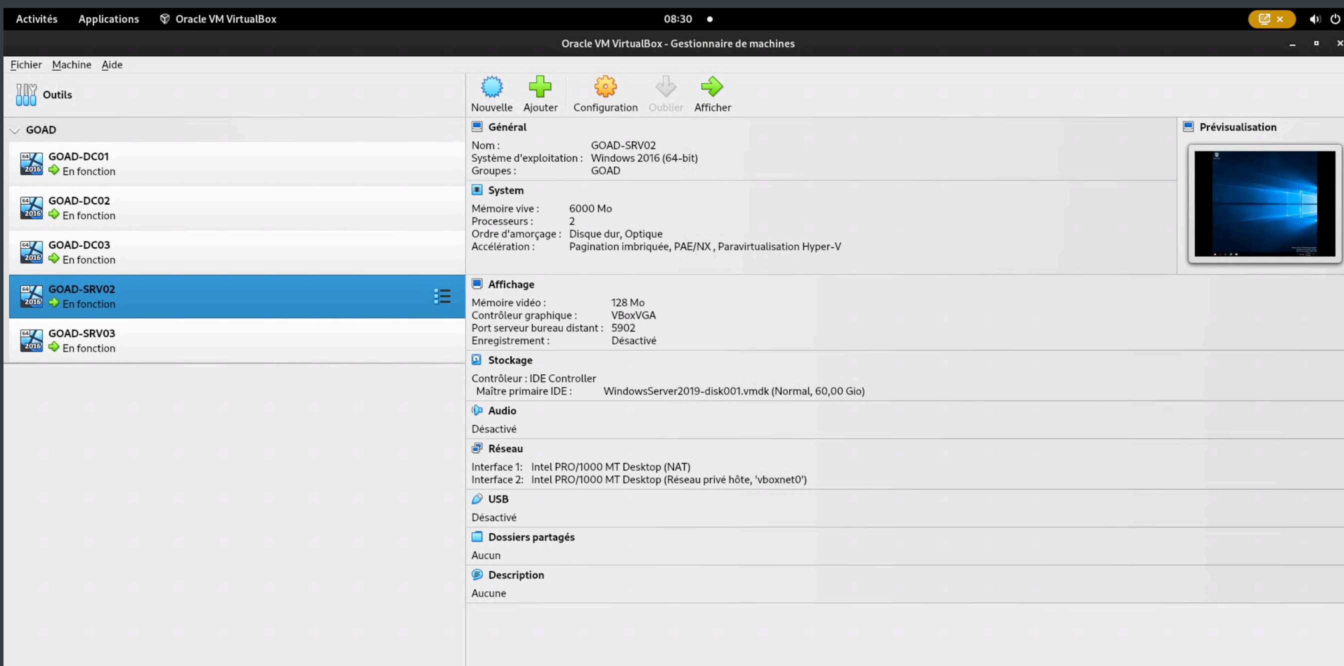
Ainsi je la corrige en faisant :

```
./goad.sh
load 870f5f-goad-virtualbox
provision_lab_from vulnerabilities.yml
```

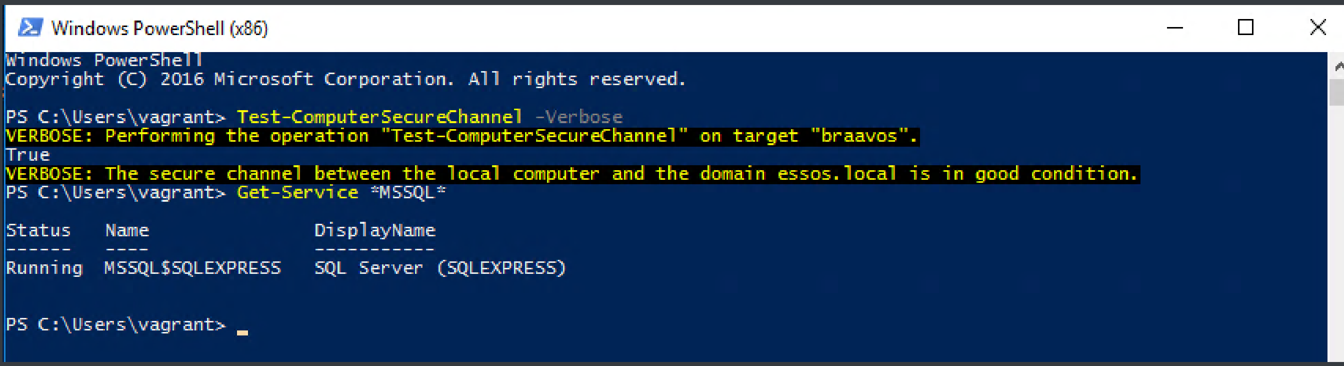
A la fin les erreurs ont bien été corrigées et tout semble bon :

```
TASK [ps : Play task ../ad/GOAD/scripts/gpo_abuse.ps1] **********************************************
changed: [dc02]
 [started TASK: ps : Play task {{ps_script}} on dc02]

TASK [ps : Play task ../ad/GOAD/scripts/rdp_scheduler.ps1] ******************************************
changed: [dc02]
 [started TASK: ps : Play task {{ps_script}} on dc03]

TASK [ps : Play task ../ad/GOAD/scripts/asrep_roasting2.ps1] ****************************************
changed: [dc03]
 [started HANDLER: vulns/adcs_esc11 : restart-adcs on srv03]

RUNNING HANDLER [vulns/adcs_esc11 : restart-adcs] ***************************************************
changed: [srv03]

PLAY RECAP ******************************************************************************************
dc01                       : ok=11   changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
dc02                       : ok=16   changed=10   unreachable=0    failed=0    skipped=6    rescued=0    ignored=0
dc03                       : ok=10   changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv02                      : ok=13   changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
srv03                      : ok=13   changed=4    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

[*] Provisioned from vulnerabilities.yml in 01:01:43

GOAD/virtualbox/docker/192.168.56.X (870f5f-goad-virtualbox) >
```

On constate bien que les VM sont apparues sur Virtualbox :

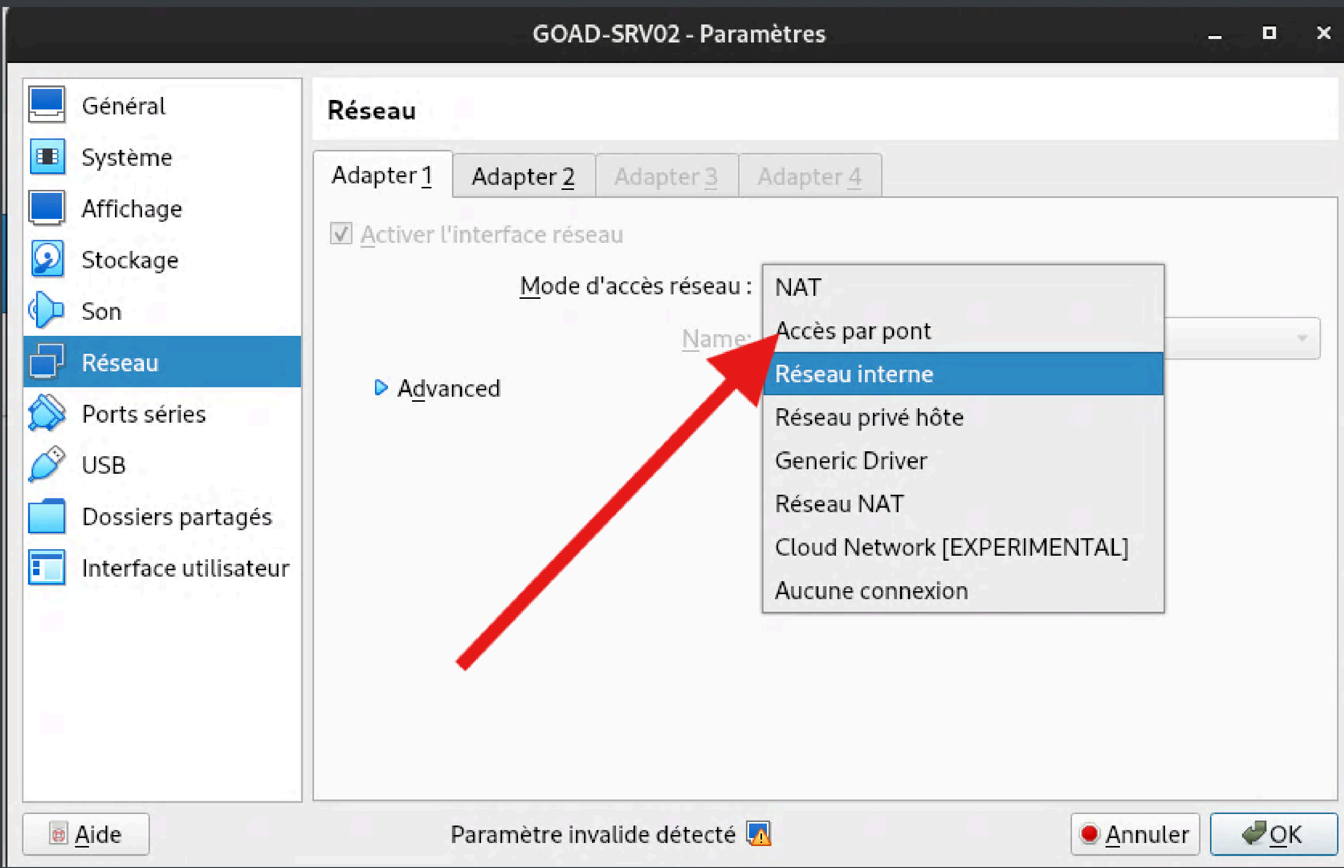


En effet quand on se connecte depuis Virtualbox sur la machine en erreur on voit bien que la connection au domaine et le service SQL Tourne :

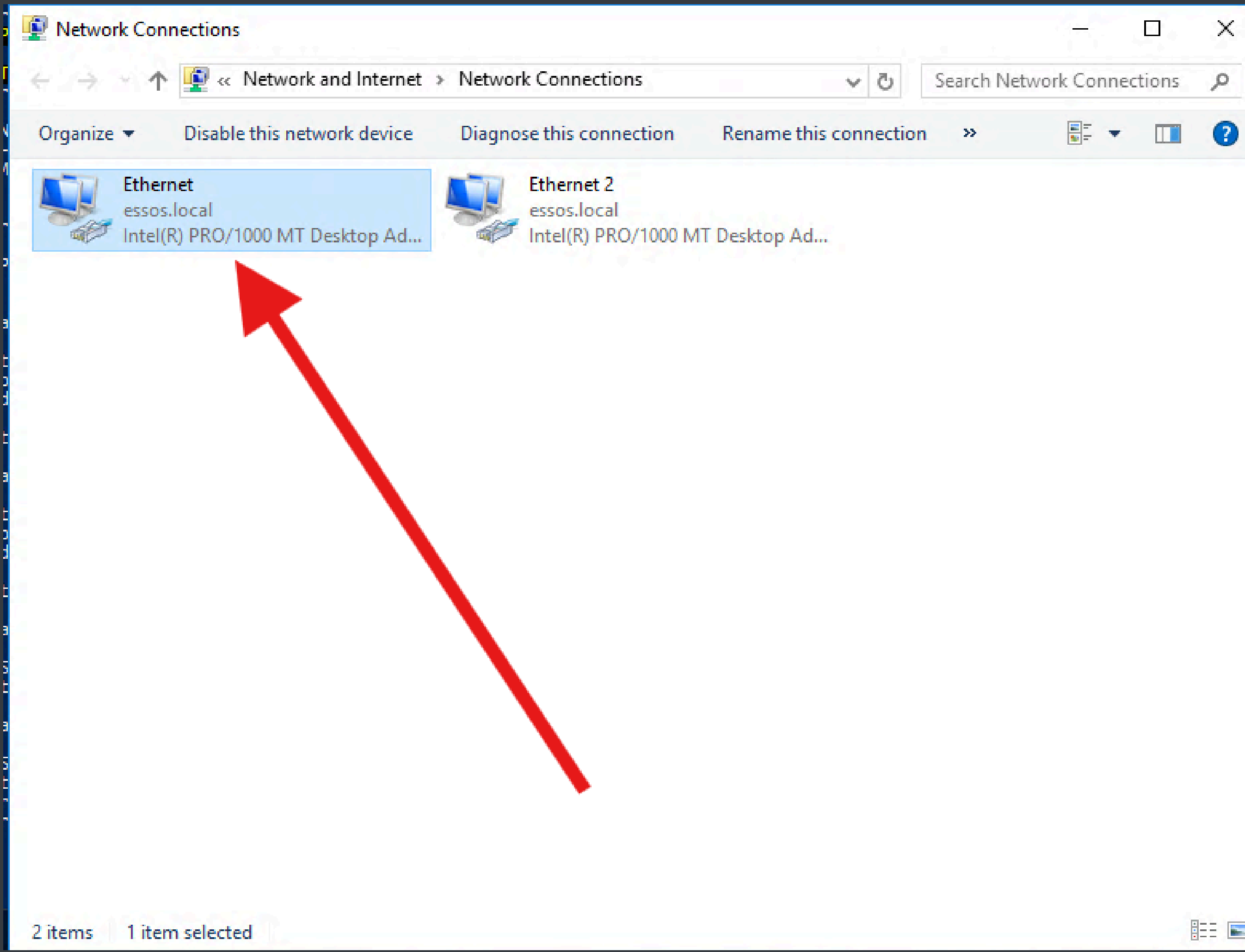


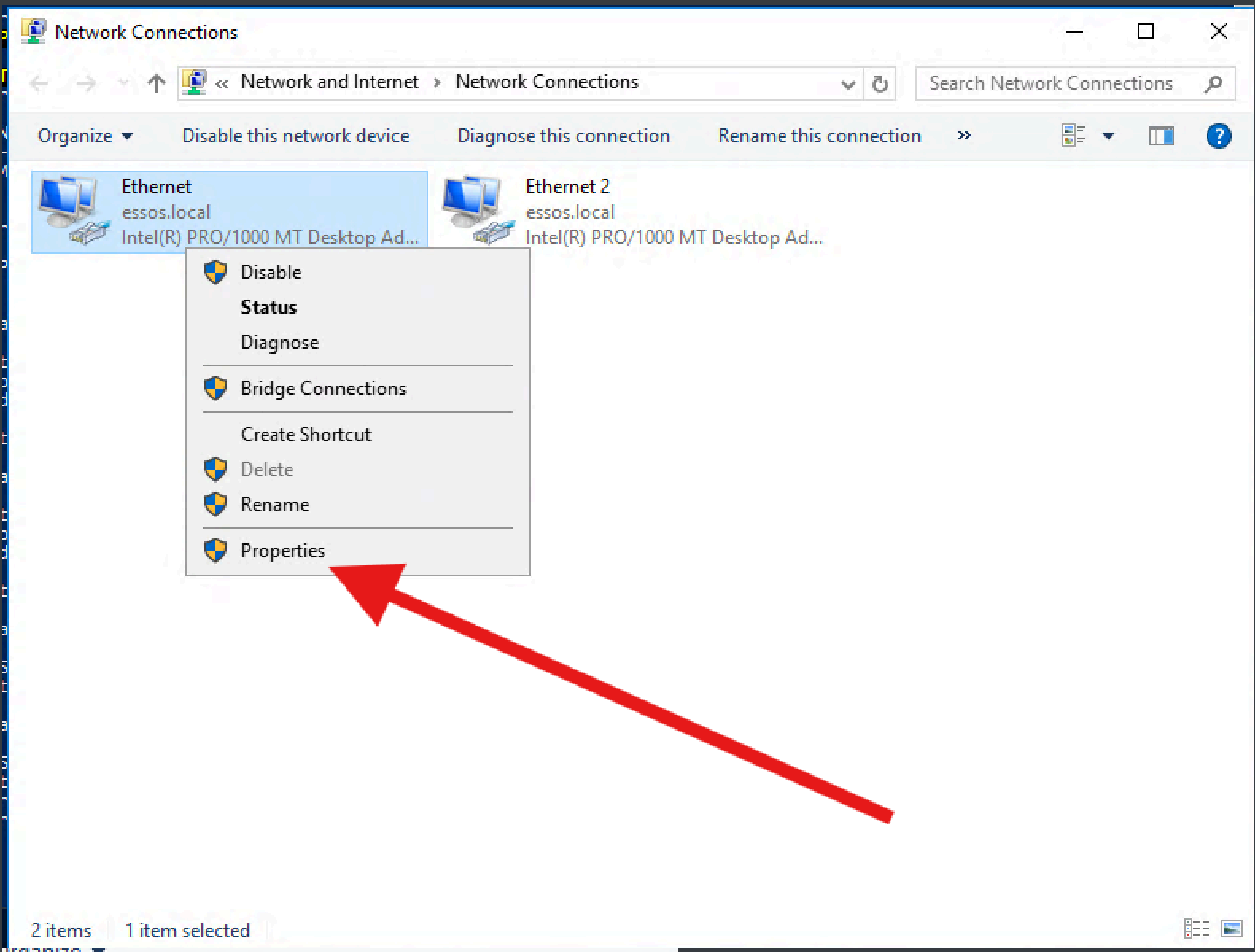
2/ Rendre les VMs accessibles sur le réseau

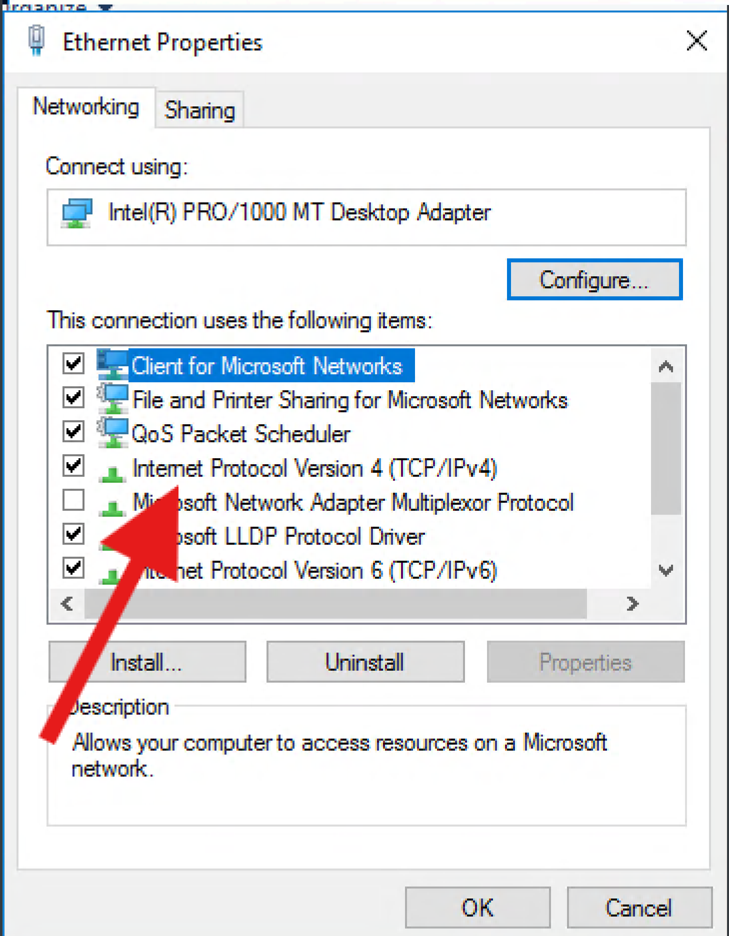
Pour chaque VM, nous allons modifier l'interface NAT par une interface pont réseau pour qu'elle soit accessible. Pour cela on se rends dans les paramètres de la VM, dans Réseau, et on passe du type NAT à pont réseau :



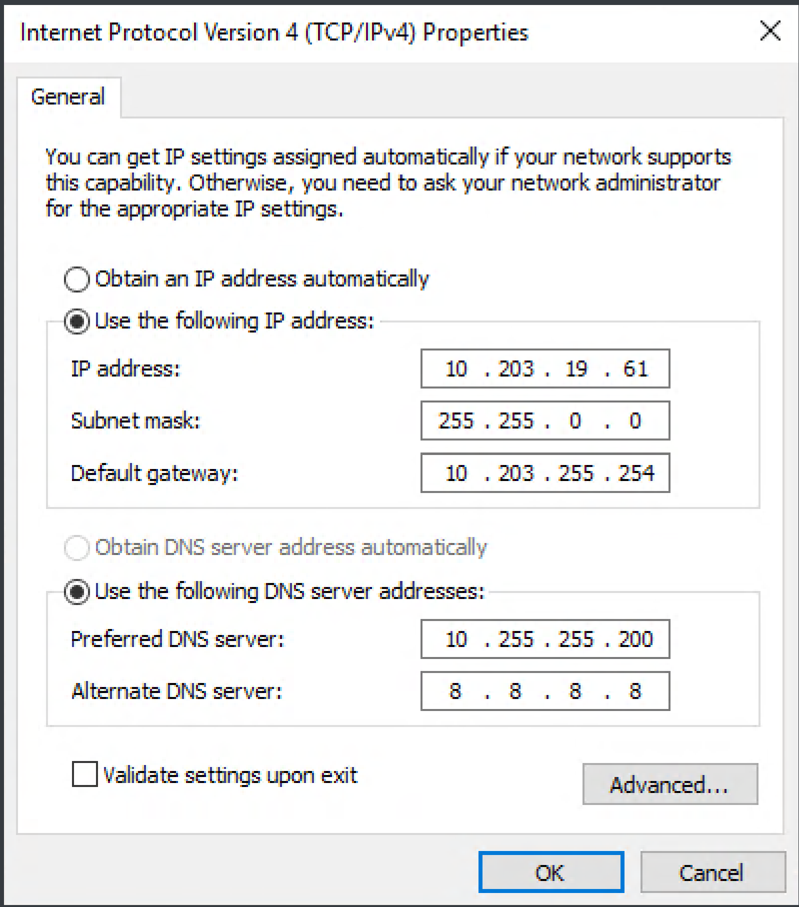
Ensuite on va forcer les IPs en dur pour qu'elles correspondent à notre plan. Pour cela on fait `ncpa.cpl` dans un terminal et on indique les adresses IP dans Ethernet (clic droit > properties > Internet Protocol V4) :



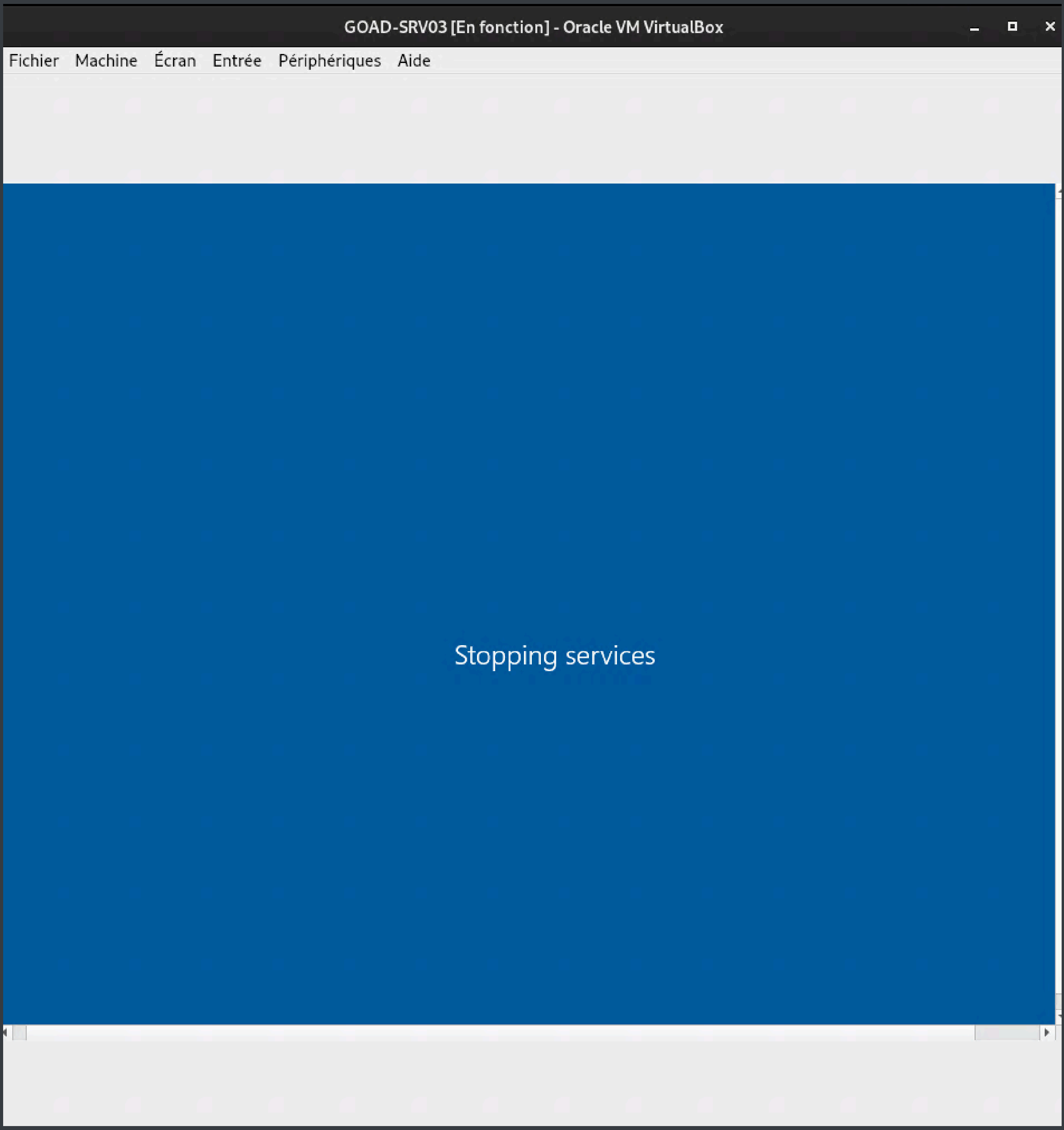




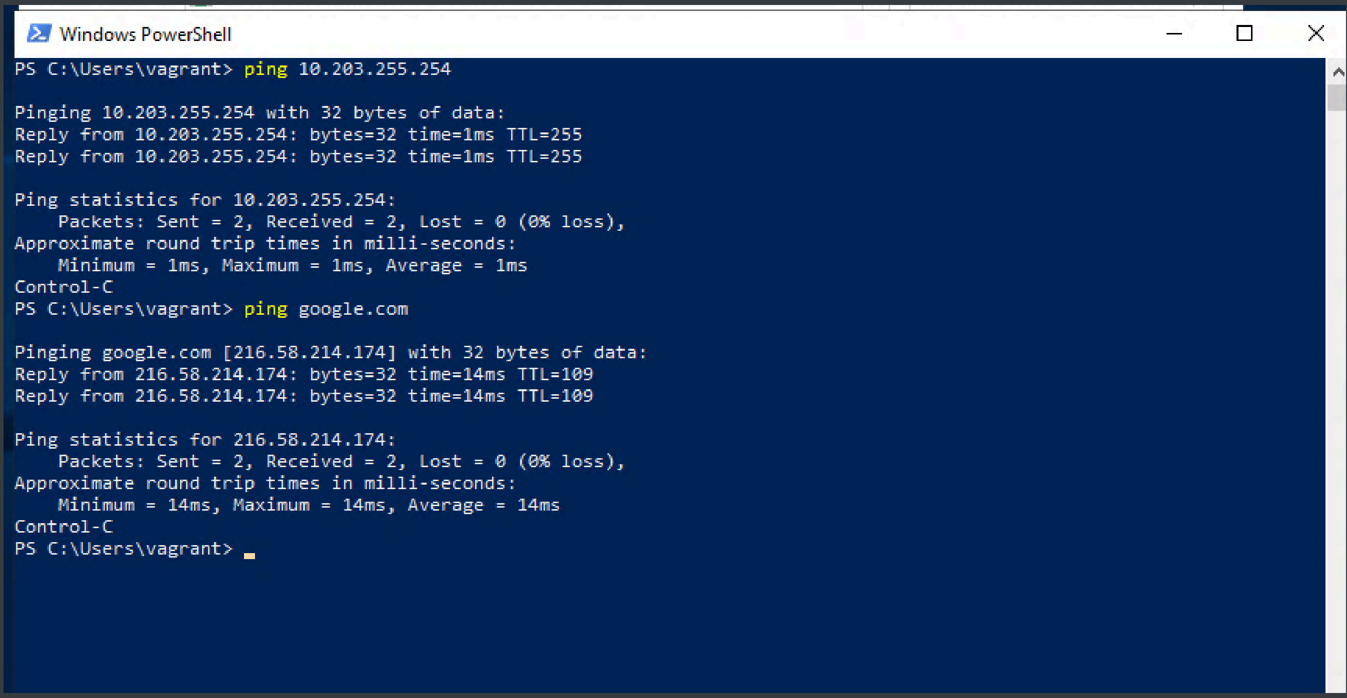
Et on renseigne les bons paramètres IPs :


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 203 . 19 . 61
Subnet mask:
255 . 255 . 0 . 0
Default gateway:
10 . 203 . 255 . 254
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
10 . 255 . 255 . 200
Alternate DNS server:
8 . 8 . 8 . 8
Validate settings upon exit
Advanced...
OK
Cancel

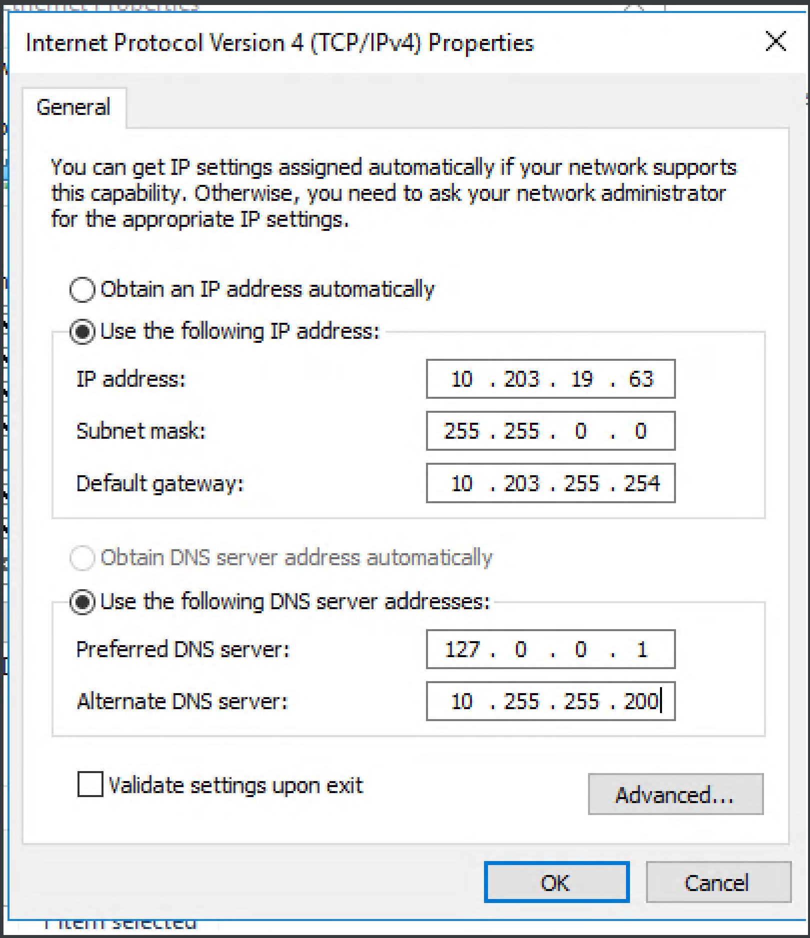
Ensuite on reboot le serveur :



Les tests nous montrent bien que le réseau fonctionne :



Les serveurs eux doivent rester leurs propres DNS primaire :



Nos VMS sont désormais prêtes.